Classical AI Reasoning System

1. Objective

The aim of this project is to build a system that decides whether a query logically follows from a knowledge base of facts and rules. The notebook converts input into a formal logical representation and uses a classical reasoning engine to produce a True or False prediction. The final decision is made by the reasoning procedures, rather than by asking a generative AI model to answer the query.

2. Representation and System Design

The system represents statements as logical formulas, including predicates, variables, facts, negation, conjunction,disjunction, implication, and quantifiers. A parser normalizes different logical notations and builds an internal syntaxtree. Rules and facts are then prepared for the appropriate inference engine. The overall pipeline is: natural-languageor formal input → normalization and parsing → logical representation → reasoning → prediction.

3. Reasoning Algorithms

The implementation combines several classical approaches. Forward chaining repeatedly applies rules to known facts, while backward chaining starts with the query and searches for rules that could prove it. Unification matches variables and terms during inference. For more general first-order logic cases, resolution is used by negating the query and checking whether the knowledge base leads to an empty clause (a contradiction). The system also selects between closed-world and open-world reasoning depending on the input type. CounterBench questions are handled with a structural causal model and the abduction–action–prediction procedure.

4. Public Dataset Results

Dataset / split	Queries	Accuracy	Time	Average time
RuleTaker test	18,000	100.00%	23.87 s	1.33 ms/query
FOLIO v2 validation	203	85.22%	2.88 s	14.20 ms/query
FOLIO v2 train	1,001	81.12%	8.20 s	8.19 ms/query
CounterBench	1,200	93.75%	0.18 s	0.15 ms/query

Combined result: Across all four reported public-data runs, the notebook evaluated 20,404 queries and reported a
weighted accuracy of **98.56%,** with a total inference time of **35.14 seconds** (about 1.72 ms per query on average).
This combined figure includes both FOLIO training and validation results, so it should be interpreted as a summary of
all reported runs rather than as a single held-out benchmark score. For a more balanced comparison across the three

main benchmark results—RuleTaker test, FOLIO validation, and CounterBench—the macro-average accuracy is
92.99%.

5. Example Reasoning Trace

Facts: Young(bob), Kind(bob)
Rules: Young(x) → Smart(x); Smart(x) ∧ Kind(x) → Successful(x)
Query: Successful(bob)
1. Young(bob) is a given fact, so the first rule derives Smart(bob).
2. Smart(bob) and Kind(bob) are now both available, so the second rule derives Successful(bob).
3. The query is therefore answered True.

6. Conclusion

The results show strong performance on the evaluated RuleTaker test questions and good performance on CounterBench, while FOLIO remains more challenging. The notebook's optional extended experiments were disabled,so the results above reflect the runs recorded in the notebook rather than a complete ablation study. The private test
file was not present when the notebook was run; therefore, no private-test accuracy or final submission.csv result is reported here. The reported accuracy should be reproduced by running the notebook in the intended environment
before submission.